

DS3001 → A2 Phase1 Problem 1

1. Show that $m(a + bX) = a + bm(X)$

$$\begin{aligned} m(a + bX) &= \frac{1}{N} \sum_{i=1}^N (a + bx_i) \\ &= \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N x_i \right) \\ &= \frac{1}{N} \left(Na + b \sum_{i=1}^N x_i \right) \\ &= a + b \left(\frac{1}{N} \sum_{i=1}^N x_i \right) \end{aligned}$$

$$\rightarrow \boxed{m(a + bX) = a + bm(X)}$$

2. Show that $\text{cov}(X, a + bY) = b \text{cov}(X, Y)$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) [(a + by_i) - m(a + bY)]$$

$$m(a + bY) = a + bm(Y)$$

$$(a + by_i) - (a + bm(Y)) = b(y_i - m(Y))$$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b(y_i - m(Y))$$

$$= b \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (y_i - m(Y))$$

$$\rightarrow \boxed{\text{cov}(X, a + bY) = b \text{cov}(X, Y)}$$

3. show that $\text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)$

$$(a + bx_i) - m(a + bX) = b(x_i - m(X))$$

$$\begin{aligned}\text{cov}(a + bX, a + bX) &= \frac{1}{N} \sum_{i=1}^N b(x_i - m(X)) b(x_i - m(X)) \\ &= b^2 \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2\end{aligned}$$

$$\Rightarrow \boxed{\text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)}$$

$$\text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$\Rightarrow \boxed{\text{cov}(X, X) = s^2}$$

4. $m(g(X)) = g(m(X)) \rightarrow$ always true???

Let $X = \{1, 3\}$ and $g(x) = x^3$

$$m(X) = \frac{1+3}{2} = 2, \quad g(m(X)) = 2^3 = 8$$

$$m(g(X)) = \frac{1^3 + 3^3}{2} = \frac{10}{2} = 5$$

$$5 \neq 8$$

$$\Rightarrow \boxed{m(g(X)) \neq g(m(X)), \text{ No}}$$